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MAE 240 HW — Problems 1 and 2

Problem 1: Barycentric N-body problem Problem 2: Bodycentric N-body problem + restricted (N+1)th body

```
clear; close all; clc;
```

Constants

```
G     = 6.67430e-20;          % km^3 kg^-1 s^-2
M_sun = 1.989e30;             % kg
AU    = 1.496e8;              % km
yr    = 365.25*24*3600;       % s
```

Bodies

```
n = 5;
body_names = {'Sun','Venus','Earth','Mars','Jupiter'};

m_bodies = [
```

```
    M_sun
    4.867e24
    5.972e24
    6.417e23
    1.898e27
];

r_AU = [0; 0.723; 1.000; 1.524; 5.203];
```

Colors

```
colors = lines(n);
sunColor = [1.0 0.75 0.0];
colors(1,:) = sunColor;

sunMarkerSize = 2.5;
```

Initial conditions

```
r0_all = zeros(3*n,1);
v0_all = zeros(3*n,1);

theta0 = linspace(0,2*pi,n+1);
theta0 = theta0(1:n);

% Sun
r0_all(1:3) = [0;0;0];
v0_all(1:3) = [0;0;0];

% Planets
for i = 2:n
    r_mag = r_AU(i)*AU;
    theta = theta0(i);

    r0_all(3*i-2:3*i) = [
        r_mag*cos(theta)
        r_mag*sin(theta)
        0
    ];

    v_c = sqrt(G*M_sun/r_mag);

    v0_all(3*i-2:3*i) = [
        -v_c*sin(theta)
         v_c*cos(theta)
         0
    ];
end

x0_bary = [r0_all; v0_all];
```

Time

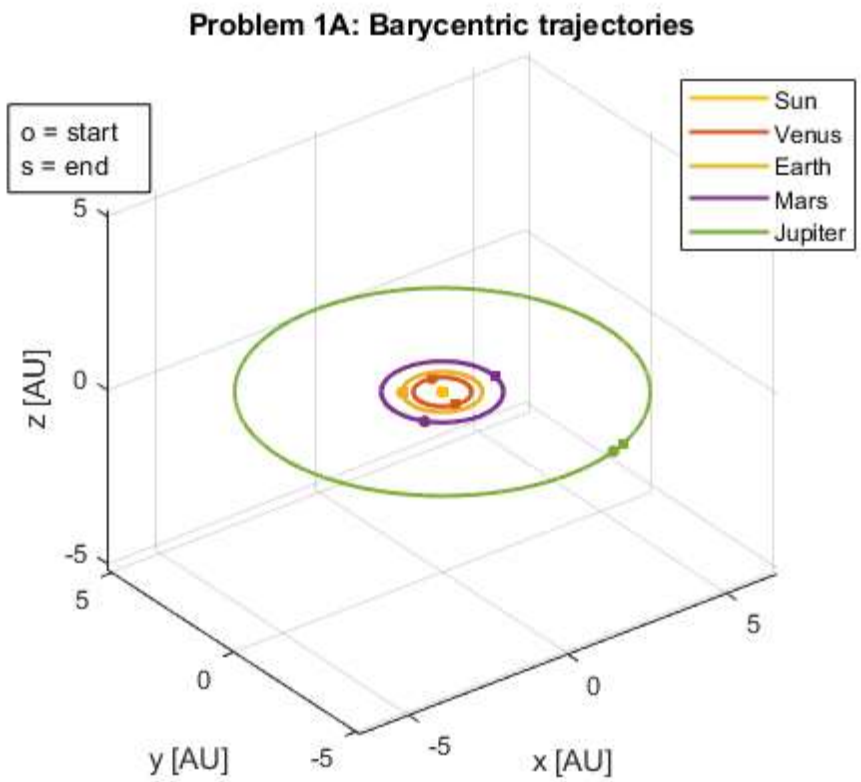
```
T_end = 12*yr;
tspan = linspace(0,T_end,5000);
opts = odeset('RelTol',1e-11,'AbsTol',1e-13);
```

PROBLEM 1A — BARYCENTRIC N-BODY

```
[T_bary,X_bary] = ode45(@(t,x) eom_NBP_bary(t,x,m_bodies,G), ...
                        tspan,x0_bary,opts);

r_bary = X_bary(:,1:3*n);
v_bary = X_bary(:,3*n+1:end);
```

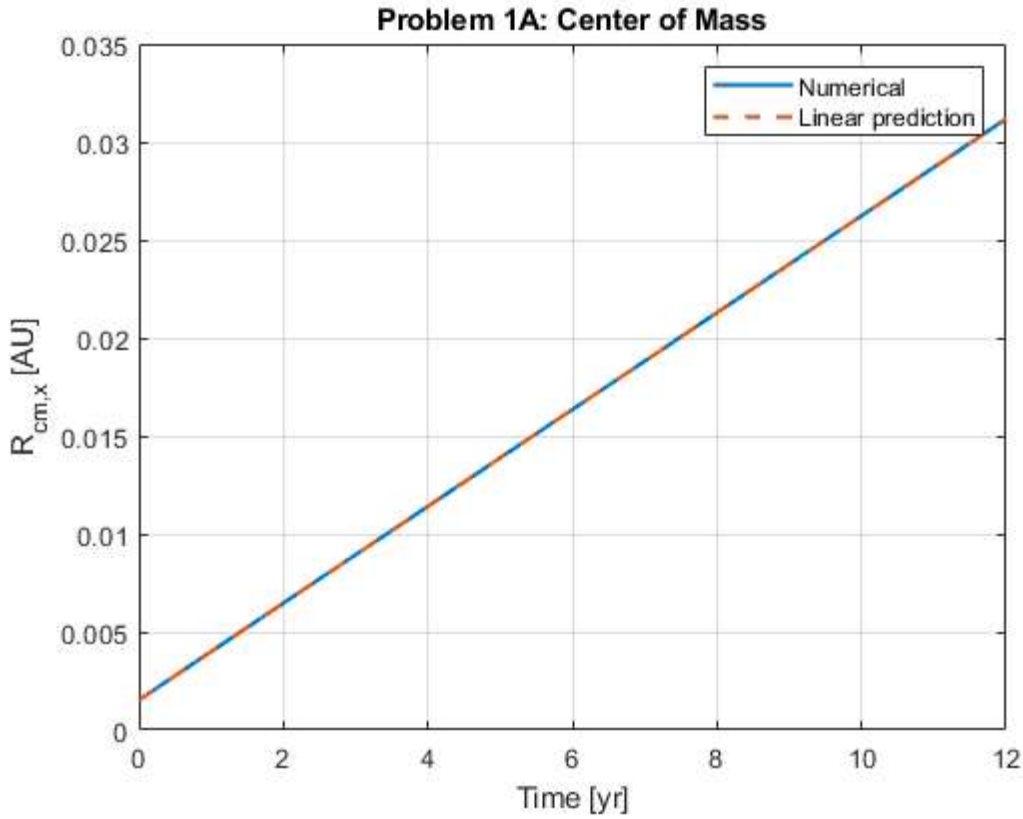
Plot Problem 1A

```
figure('Color','w','Name','Problem 1A — Barycentric');
hold on; grid on; axis equal;

for i = 1:n
    ri = r_bary(:,3*i-2:3*i);

    plot3(ri(:,1)/AU,ri(:,2)/AU,ri(:,3)/AU, ...
        'LineWidth',1.5, ...
        'Color',colors(i,:), ...
        'DisplayName',body_names{i});

    plot3(ri(1,1)/AU,ri(1,2)/AU,ri(1,3)/AU,'o', ...
        'MarkerSize',4, ...
        'Color',colors(i,:), ...
        'MarkerFaceColor',colors(i,:), ...
        'HandleVisibility','off');

    plot3(ri(end,1)/AU,ri(end,2)/AU,ri(end,3)/AU,'s', ...
        'MarkerSize',4, ...
        'Color',colors(i,:), ...
        'MarkerFaceColor',colors(i,:), ...
        'HandleVisibility','off');
end

xlabel('x [AU]');
ylabel('y [AU]');
zlabel('z [AU]');
title('Problem 1A: Barycentric trajectories');
legend('Location','best');
view(3);

annotation('textbox',[0.13 0.78 0.18 0.08], ...
    'String',{'o = start','s = end'}, ...
    'FitBoxToText','on', ...
    'BackgroundColor','w');
```



Problem 1A — Barycenter

```
r_cm = centerOfMassHistory(r_bary,m_bodies);
P = linearMomentumHistory(v_bary,m_bodies);
V_cm0 = P(1,:)/sum(m_bodies);
r_cm_linear = r_cm(1,:) + T_bary(:)*V_cm0;

figure('Color','w','Name','Problem 1A — Center of Mass');
plot(T_bary/yr,r_cm(:,1)/AU,'LineWidth',1.5); hold on;
plot(T_bary/yr,r_cm_linear(:,1)/AU,'--','LineWidth',1.5);
grid on;
xlabel('Time [yr]');
ylabel('R_{cm,x} [AU]');
legend('Numerical','Linear prediction');
title('Problem 1A: Center of Mass');
```



Problem 1A — Energy and Angular Momentum

```
[~,~,E] = totalEnergyHistory(r_bary,v_bary,m_bodies,G);
relE = abs(E-E(1))/abs(E(1));

H_bary = angularMomentumHistory(r_bary,v_bary,m_bodies);
relH_bary = vecnorm(H_bary-H_bary(1,:),2,2)/norm(H_bary(1,:));
```

PROBLEM 1B — RESTRICTED BODY IN BARYCENTRIC FRAME

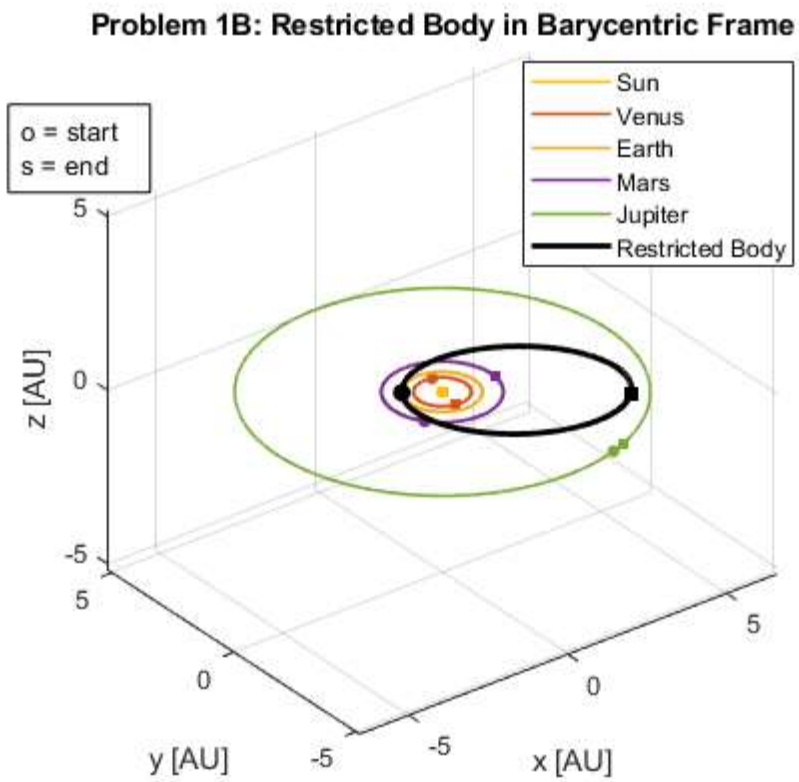
```
iEarth = 3;
iJupiter = 5;

r0_earth_bary = r_bary(1,3*iEarth-2:3*iEarth).';
v0_earth_bary = v_bary(1,3*iEarth-2:3*iEarth).';

r0_jup_bary = r_bary(1,3*iJupiter-2:3*iJupiter).';

u_r = r0_earth_bary/norm(r0_earth_bary);
u_t = v0_earth_bary/norm(v0_earth_bary);

offset = 1e-3*AU;
r0_sc_bary = r0_earth_bary + offset*u_r;

a_trans = 0.5*(norm(r0_sc_bary) + norm(r0_jup_bary));
v_dep = sqrt(G*M_sun*(2/norm(r0_sc_bary) - 1/a_trans));

v0_sc_bary = v_dep*u_t;
x0_sc_bary = [r0_sc_bary; v0_sc_bary];
```

```
r_interp_bary = makeInterpolants(T_bary,r_bary,n);

[T_sc_bary,X_sc_bary] = ode45(@(t,x) eom_sc_barycentric(t,x,r_interp_bary,m_bodies,G), ...
                              tspan,x0_sc_bary,opts);

r_sc_bary = X_sc_bary(:,1:3);
v_sc_bary = X_sc_bary(:,4:6);
```

Plot Problem 1B

```
figure('Color','w','Name','Problem 1B — Restricted Body Barycentric');
hold on; grid on; axis equal;

for i = 1:n
    ri = r_bary(:,3*i-2:3*i);

    plot3(ri(:,1)/AU,ri(:,2)/AU,ri(:,3)/AU, ...
        'LineWidth',1.2, ...
        'Color',colors(i,:), ...
        'DisplayName',body_names{i});

    plot3(ri(1,1)/AU,ri(1,2)/AU,ri(1,3)/AU,'o', ...
        'MarkerSize',4, ...
        'Color',colors(i,:), ...
        'MarkerFaceColor',colors(i,:), ...
        'HandleVisibility','off');

    plot3(ri(end,1)/AU,ri(end,2)/AU,ri(end,3)/AU,'s', ...
        'MarkerSize',4, ...
        'Color',colors(i,:), ...
        'MarkerFaceColor',colors(i,:), ...
        'HandleVisibility','off');
end

plot3(r_sc_bary(:,1)/AU,r_sc_bary(:,2)/AU,r_sc_bary(:,3)/AU, ...
    'k','LineWidth',2,'DisplayName','Restricted Body');

plot3(r_sc_bary(1,1)/AU,r_sc_bary(1,2)/AU,r_sc_bary(1,3)/AU,'ko', ...
    'MarkerSize',6, ...
    'MarkerFaceColor','k', ...
    'HandleVisibility','off');

plot3(r_sc_bary(end,1)/AU,r_sc_bary(end,2)/AU,r_sc_bary(end,3)/AU,'ks', ...
    'MarkerSize',6, ...
    'MarkerFaceColor','k', ...
    'HandleVisibility','off');

xlabel('x [AU]');
ylabel('y [AU]');
zlabel('z [AU]');
title('Problem 1B: Restricted Body in Barycentric Frame');
legend('Location','best');
view(3);

annotation('textbox',[0.13 0.78 0.18 0.08], ...
    'String',{'o = start','s = end'}, ...
```

```
    'FitBoxToText','on', ...
    'BackgroundColor','w');
```



Problem 1B — Energy and angular momentum diagnostics

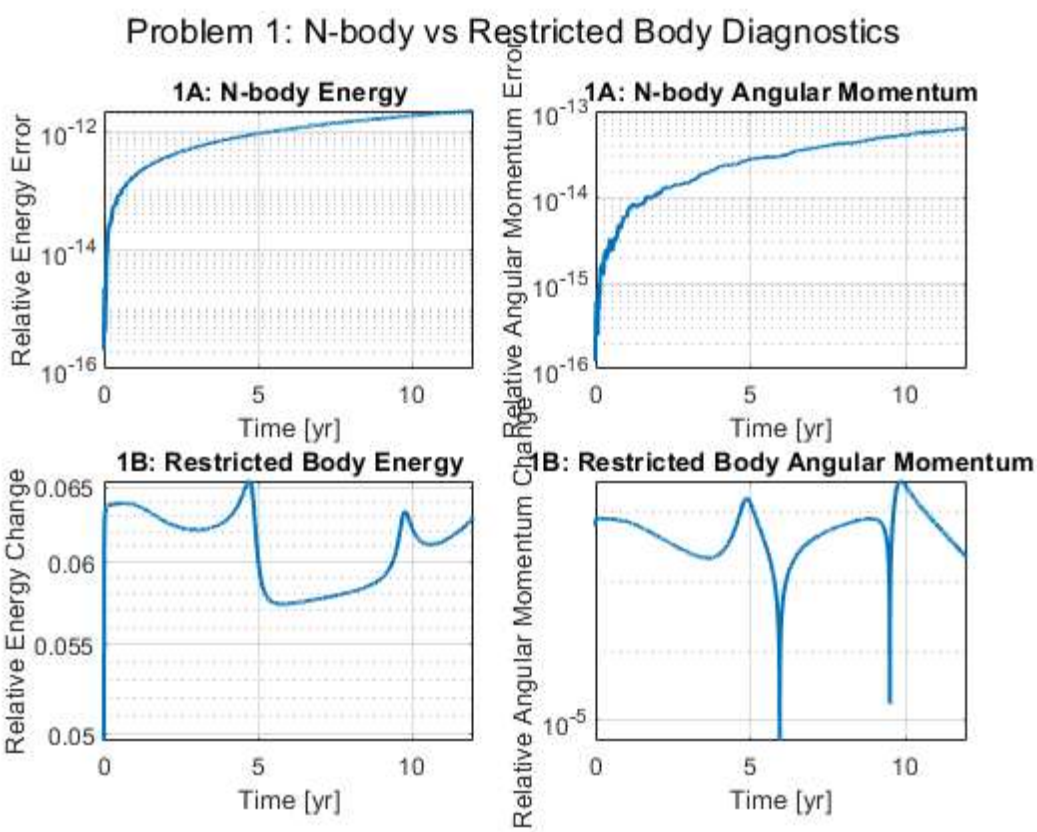
```
E_sc_bary = restrictedEnergyHistory(T_sc_bary,r_sc_bary,v_sc_bary,r_interp_bary,m_bodies,G);
H_sc_bary = cross(r_sc_bary,v_sc_bary,2);

relE_sc_bary = abs(E_sc_bary-E_sc_bary(1))/abs(E_sc_bary(1));
relH_sc_bary = vecnorm(H_sc_bary-H_sc_bary(1,:),2,2)/norm(H_sc_bary(1,:));
```

Problem 1 — Combined Energy and Angular Momentum Diagnostics

```
figure('Color','w','Name','Problem 1 — Energy and Angular Momentum Diagnostics');

subplot(2,2,1);
semilogy(T_bary/yr,relE,'LineWidth',1.5);
grid on;
xlabel('Time [yr]');
ylabel('Relative Energy Error');
title('1A: N-body Energy');

subplot(2,2,2);
semilogy(T_bary/yr,relH_bary,'LineWidth',1.5);
grid on;
xlabel('Time [yr]');
ylabel('Relative Angular Momentum Error');
title('1A: N-body Angular Momentum');
```

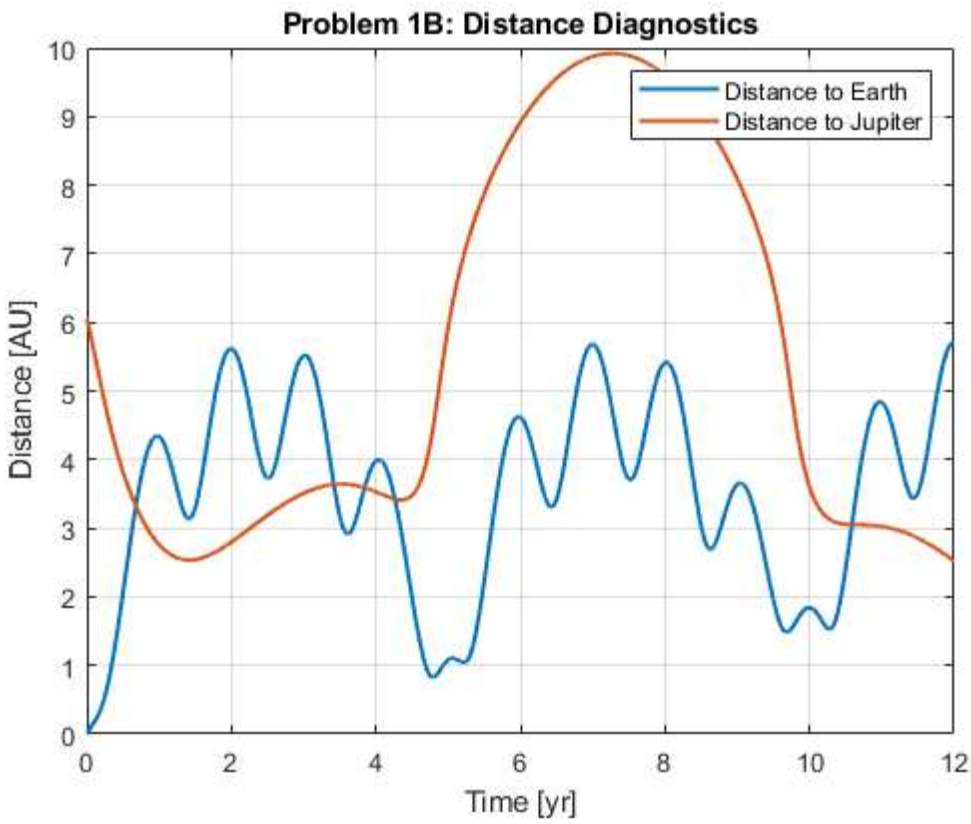
```
subplot(2,2,3);
semilogy(T_sc_bary/yr,relE_sc_bary,'LineWidth',1.5);
grid on;
xlabel('Time [yr]');
ylabel('Relative Energy Change');
title('1B: Restricted Body Energy');

subplot(2,2,4);
semilogy(T_sc_bary/yr,relH_sc_bary,'LineWidth',1.5);
grid on;
xlabel('Time [yr]');
ylabel('Relative Angular Momentum Change');
title('1B: Restricted Body Angular Momentum');

sgtitle('Problem 1: N-body vs Restricted Body Diagnostics');

fprintf(['Figure 4 shows that the full N-body system conserves energy and angular \n' ...
    'momentum very well, with only small numerical error. In contrast, the \n' ...
    'restricted body does not conserve energy or angular momentum because it moves \n' ...
    'through the time-varying gravitational field of the moving planets, while its \n' ...
    'own mass does not affect the system.\n']);
```

```
Figure 4 shows that the full N-body system conserves energy and angular
momentum very well, with only small numerical error. In contrast, the
restricted body does not conserve energy or angular momentum because it moves
through the time-varying gravitational field of the moving planets, while its
own mass does not affect the system.
```



Problem 1B — Distance diagnostics

```
rE_bary = zeros(length(T_sc_bary),3);
rJ_bary = zeros(length(T_sc_bary),3);

for k = 1:length(T_sc_bary)
    rE_bary(k,:) = r_interp_bary{iEarth}(T_sc_bary(k)).';
    rJ_bary(k,:) = r_interp_bary{iJupiter}(T_sc_bary(k)).';
end

dE_bary = vecnorm(r_sc_bary-rE_bary,2,2);
dJ_bary = vecnorm(r_sc_bary-rJ_bary,2,2);

figure('Color','w','Name','Problem 1B — Distance Diagnostics');
plot(T_sc_bary/yr,dE_bary/AU,'LineWidth',1.5); hold on;
plot(T_sc_bary/yr,dJ_bary/AU,'LineWidth',1.5);
grid on;
xlabel('Time [yr]');
ylabel('Distance [AU]');
legend('Distance to Earth','Distance to Jupiter');
title('Problem 1B: Distance Diagnostics');
```



PROBLEM 2A — BODYCENTRIC N-BODY

```
r0_body = zeros(3*n,1);
v0_body = zeros(3*n,1);

for i = 1:n
    r0_body(3*i-2:3*i) = r0_all(3*i-2:3*i) - r0_all(1:3);
    v0_body(3*i-2:3*i) = v0_all(3*i-2:3*i) - v0_all(1:3);
end
```

```
x0_body = [r0_body; v0_body];

[T_body,X_body] = ode45(@(t,x) eom_NBP_bodycentric(t,x,m_bodies,G), ...
                        tspan,x0_body,opts);

r_body = X_body(:,1:3*n);
v_body = X_body(:,3*n+1:end);
```

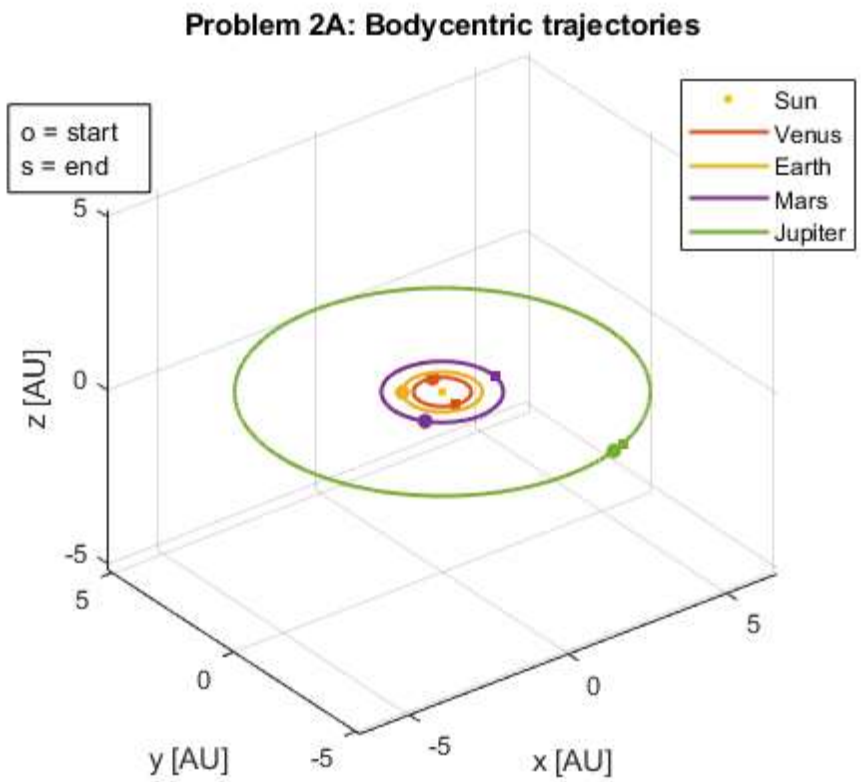
Convert barycentric solution to Sun-centered form

```
r_bary_rel = zeros(size(r_bary));

for k = 1:length(T_bary)
    rSun = r_bary(k,1:3);

    for i = 1:n
        r_bary_rel(k,3*i-2:3*i) = r_bary(k,3*i-2:3*i) - rSun;
    end
end
```

Plot Problem 2A

```
figure('Color','w','Name','Problem 2A — Bodycentric');
hold on; grid on; axis equal;

for i = 1:n
    ri = r_body(:,3*i-2:3*i);

    if i == 1
        plot3(0,0,0,'o', ...
            'MarkerSize',sunMarkerSize, ...
            'MarkerFaceColor',sunColor, ...
            'MarkerEdgeColor',sunColor, ...
            'DisplayName','Sun');
    else
        plot3(ri(:,1)/AU,ri(:,2)/AU,ri(:,3)/AU, ...
            'LineWidth',1.5, ...
            'Color',colors(i,:), ...
            'DisplayName',body_names{i});

        plot3(ri(1,1)/AU,ri(1,2)/AU,ri(1,3)/AU,'o', ...
            'MarkerSize',5, ...
            'Color',colors(i,:), ...
            'MarkerFaceColor',colors(i,:), ...
            'HandleVisibility','off');

        plot3(ri(end,1)/AU,ri(end,2)/AU,ri(end,3)/AU,'s', ...
            'MarkerSize',5, ...
            'Color',colors(i,:), ...
            'MarkerFaceColor',colors(i,:), ...
            'HandleVisibility','off');
    end
end

xlabel('x [AU]');
```

```
ylabel('y [AU]');
zlabel('z [AU]');
title('Problem 2A: Bodycentric trajectories');
legend('Location','best');
view(3);

annotation('textbox',[0.13 0.78 0.18 0.08], ...
    'String',{'o = start','s = end'}, ...
    'FitBoxToText','on', ...
    'BackgroundColor','w');
```

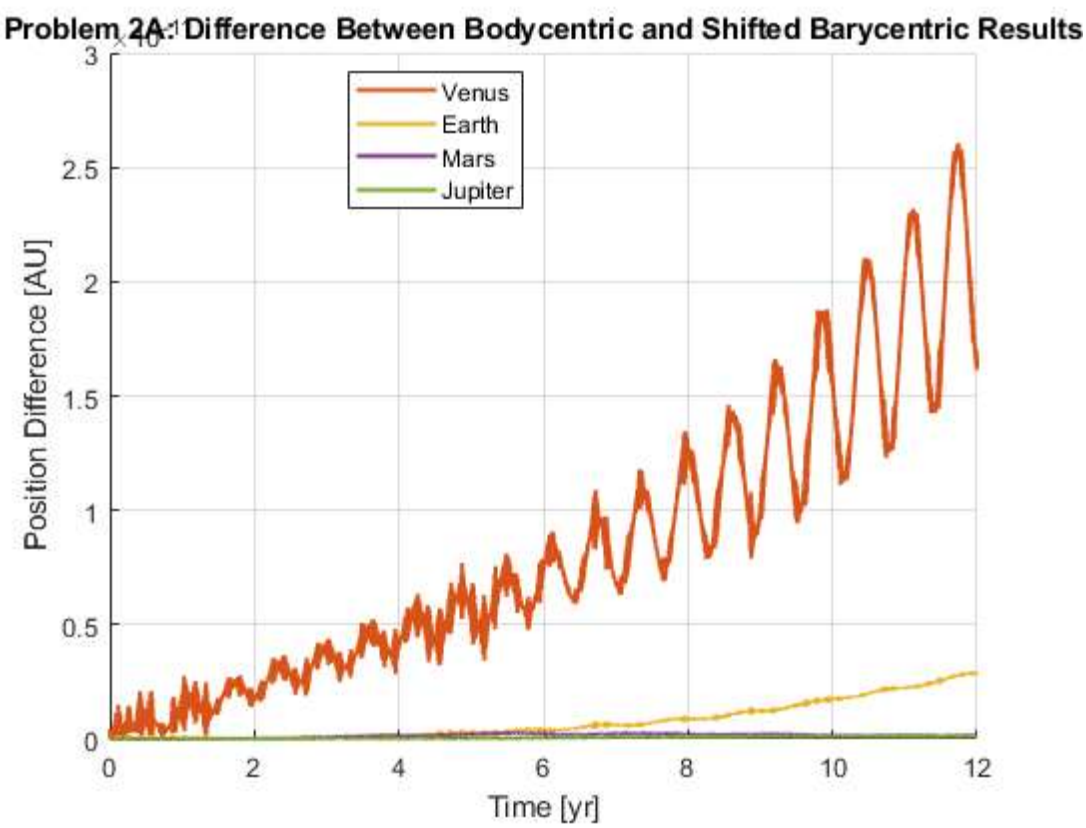


Problem 2A — Integration accuracy comparison error

```
figure('Color','w','Name','Problem 2A — Bodycentric Comparison Error');
hold on; grid on;

for i = 2:n
    ri_body = r_body(:,3*i-2:3*i);
    ri_bary_rel = r_bary_rel(:,3*i-2:3*i);

    err_i = vecnorm(ri_body-ri_bary_rel,2,2);

    semilogy(T_body/yr,err_i/AU, ...
        'LineWidth',1.5, ...
        'Color',colors(i,:), ...
        'DisplayName',body_names{i});
end

xlabel('Time [yr]');
ylabel('Position Difference [AU]');
title('Problem 2A: Difference Between Bodycentric and Shifted Barycentric Results');
```

```
legend('Location','best');

fprintf(['\nFigure 7 compares the direct bodycentric solution with the barycentric \n' ...
    'solution shifted into the Sun-centered frame. The errors remain small, showing \n' ...
    'that the bodycentric formulation is working correctly. Venus has the largest \n' ...
    'error because it is closest to the Sun and the system center of mass, so small\n' ...
    'numerical or coordinate differences are more noticeable in its trajectory.\n']);
```

```
Figure 7 compares the direct bodycentric solution with the barycentric
solution shifted into the Sun-centered frame. The errors remain small, showing
that the bodycentric formulation is working correctly. Venus has the largest
error because it is closest to the Sun and the system center of mass, so small
numerical or coordinate differences are more noticeable in its trajectory.
```



PROBLEM 2B — RESTRICTED BODY IN BODYCENTRIC FRAME

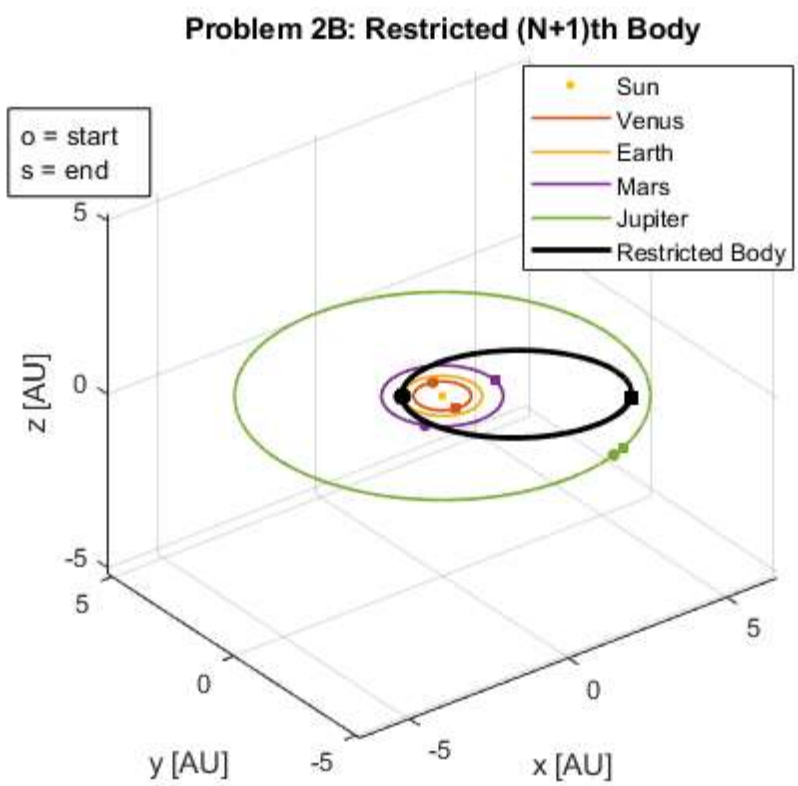
```
r0_earth = r_body(1,3*iEarth-2:3*iEarth).';
v0_earth = v_body(1,3*iEarth-2:3*iEarth).';
r0_jup   = r_body(1,3*iJupiter-2:3*iJupiter).';

u_r = r0_earth/norm(r0_earth);
u_t = v0_earth/norm(v0_earth);

offset = 1e-3*AU;
r0_sc = r0_earth + offset*u_r;

a_trans = 0.5*(norm(r0_sc) + norm(r0_jup));
v_dep = sqrt(G*M_sun*(2/norm(r0_sc) - 1/a_trans));
```

```
x0_sc = [r0_sc; v_dep*u_t];

r_interp = makeInterpolants(T_body,r_body,n);

[T_sc,X_sc] = ode45(@(t,x) eom_sc_bodycentric(t,x,r_interp,m_bodies,G), ...
                    tspan,x0_sc,opts);

r_sc = X_sc(:,1:3);
```

Plot Problem 2B

```
figure('Color','w','Name','Problem 2B — Restricted Body');
hold on; grid on; axis equal;

for i = 1:n
    ri = r_body(:,3*i-2:3*i);

    if i == 1
        plot3(0,0,0,'o', ...
            'MarkerSize',sunMarkerSize, ...
            'MarkerFaceColor',sunColor, ...
            'MarkerEdgeColor',sunColor, ...
            'DisplayName','Sun');

    else
        plot3(ri(:,1)/AU,ri(:,2)/AU,ri(:,3)/AU, ...
            'LineWidth',1.2, ...
            'Color',colors(i,:), ...
            'DisplayName',body_names{i});

        plot3(ri(1,1)/AU,ri(1,2)/AU,ri(1,3)/AU,'o', ...
            'MarkerSize',4, ...
            'Color',colors(i,:), ...
            'MarkerFaceColor',colors(i,:), ...
            'HandleVisibility','off');

        plot3(ri(end,1)/AU,ri(end,2)/AU,ri(end,3)/AU,'s', ...
            'MarkerSize',4, ...
            'Color',colors(i,:), ...
            'MarkerFaceColor',colors(i,:), ...
            'HandleVisibility','off');
    end
end

plot3(r_sc(:,1)/AU,r_sc(:,2)/AU,r_sc(:,3)/AU, ...
    'k','LineWidth',2,'DisplayName','Restricted Body');

plot3(r_sc(1,1)/AU,r_sc(1,2)/AU,r_sc(1,3)/AU,'ko', ...
    'MarkerSize',6, ...
    'MarkerFaceColor','k', ...
    'HandleVisibility','off');

plot3(r_sc(end,1)/AU,r_sc(end,2)/AU,r_sc(end,3)/AU,'ks', ...
    'MarkerSize',6, ...
    'MarkerFaceColor','k', ...
    'HandleVisibility','off');
```

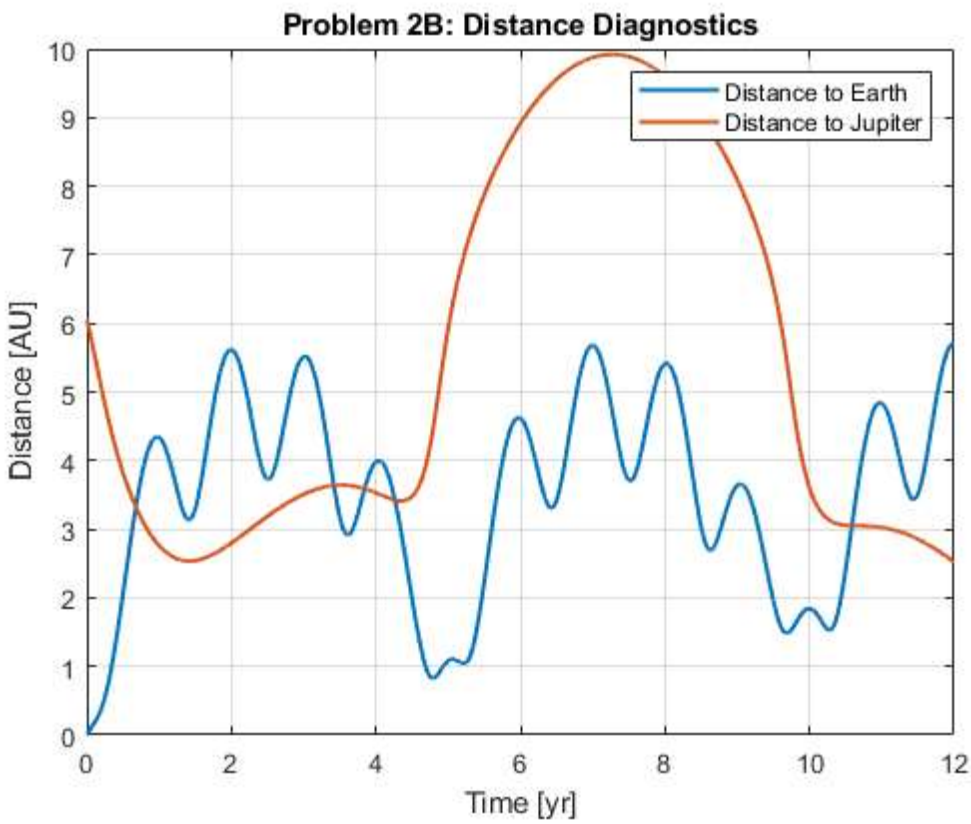
```
xlabel('x [AU]');
ylabel('y [AU]');
zlabel('z [AU]');
title('Problem 2B: Restricted (N+1)th Body');
legend('Location','best');
view(3);

annotation('textbox',[0.13 0.78 0.18 0.08], ...
    'String',{'o = start','s = end'}, ...
    'FitBoxToText','on', ...
    'BackgroundColor','w');
```



Problem 2B — Distance diagnostics

```
rE = zeros(length(T_sc),3);
rJ = zeros(length(T_sc),3);

for k = 1:length(T_sc)
    rE(k,:) = r_interp{iEarth}(T_sc(k)).';
    rJ(k,:) = r_interp{iJupiter}(T_sc(k)).';
end

dE = vecnorm(r_sc-rE,2,2);
dJ = vecnorm(r_sc-rJ,2,2);

figure('Color','w','Name','Problem 2B — Distance Diagnostics');
plot(T_sc/yr,dE/AU,'LineWidth',1.5); hold on;
plot(T_sc/yr,dJ/AU,'LineWidth',1.5);
grid on;
xlabel('Time [yr]');
ylabel('Distance [AU]');
```

```
legend('Distance to Earth','Distance to Jupiter');
title('Problem 2B: Distance Diagnostics');
```



FUNCTIONS

```
function dx = eom_NBP_bary(~,x,m,G)

n = numel(m);
r = x(1:3*n);
v = x(3*n+1:end);
a = zeros(3*n,1);

for i = 1:n
    ri = r(3*i-2:3*i);
    ai = zeros(3,1);

    for j = 1:n
        if j == i
            continue;
        end

        rj = r(3*j-2:3*j);
        rij = ri-rj;

        ai = ai - G*m(j)*rij/norm(rij)^3;
    end

    a(3*i-2:3*i) = ai;
end

dx = [v; a];
```

```
end

function dx = eom_NBP_bodycentric(~,x,m,G)

n = numel(m);
r = x(1:3*n);
v = x(3*n+1:end);
a = zeros(3*n,1);

a(1:3) = [0;0;0];

for i = 2:n
    ri = r(3*i-2:3*i);

    ai = -G*(m(1)+m(i))*ri/norm(ri)^3;

    for j = 2:n
        if j == i
            continue;
        end

        rj = r(3*j-2:3*j);
        rij = ri-rj;

        ai = ai - G*m(j)*(rij/norm(rij)^3 + rj/norm(rj)^3);
    end

    a(3*i-2:3*i) = ai;
end

dx = [v; a];
end

function dx = eom_sc_barycentric(t,x,r_interp,m,G)

n = numel(m);

r = x(1:3);
v = x(4:6);

a = zeros(3,1);

for j = 1:n
    rj = r_interp{j}(t);
    rj = rj(:);

    rij = r-rj;

    a = a - G*m(j)*rij/norm(rij)^3;
end

dx = [v; a];
end

function dx = eom_sc_bodycentric(t,x,r_interp,m,G)

n = numel(m);
```

```
r = x(1:3);
v = x(4:6);

a = -G*m(1)*r/norm(r)^3;

for j = 2:n
    rj = r_interp{j}(t);
    rj = rj(:);

    rij = r-rj;

    a = a - G*m(j)*(rij/norm(rij)^3 + rj/norm(rj)^3);
end

dx = [v; a];
end

function rcm = centerOfMassHistory(r_all,m)

n = numel(m);
Nt = size(r_all,1);
M = sum(m);

rcm = zeros(Nt,3);

for i = 1:n
    rcm = rcm + m(i)*r_all(:,3*i-2:3*i);
end

rcm = rcm/M;
end

function P = linearMomentumHistory(v_all,m)

n = numel(m);
Nt = size(v_all,1);

P = zeros(Nt,3);

for i = 1:n
    P = P + m(i)*v_all(:,3*i-2:3*i);
end
end

function H = angularMomentumHistory(r_all,v_all,m)

n = numel(m);
Nt = size(r_all,1);

H = zeros(Nt,3);

for k = 1:Nt
    Hk = zeros(1,3);

    for i = 1:n
        ri = r_all(k,3*i-2:3*i);
        vi = v_all(k,3*i-2:3*i);
```

```
        Hk = Hk + cross(ri,m(i)*vi);
    end

    H(k,:) = Hk;
end
end

function [KE,PE,E] = totalEnergyHistory(r_all,v_all,m,G)

n = numel(m);
Nt = size(r_all,1);

KE = zeros(Nt,1);
PE = zeros(Nt,1);

for k = 1:Nt
    for i = 1:n
        vi = v_all(k,3*i-2:3*i);
        KE(k) = KE(k) + 0.5*m(i)*dot(vi,vi);
    end

    for i = 1:n-1
        ri = r_all(k,3*i-2:3*i);

        for j = i+1:n
            rj = r_all(k,3*j-2:3*j);
            PE(k) = PE(k) - G*m(i)*m(j)/norm(ri-rj);
        end
    end
end

E = KE + PE;
end

function E = restrictedEnergyHistory(T,r_sc,v_sc,r_interp,m,G)

Nt = length(T);
n = numel(m);
E = zeros(Nt,1);

for k = 1:Nt
    r = r_sc(k,:).';
    v = v_sc(k,:).';

    KE = 0.5*dot(v,v);
    PE = 0;

    for j = 1:n
        rj = r_interp{j}(T(k));
        rj = rj(:);

        PE = PE - G*m(j)/norm(r-rj);
    end

    E(k) = KE + PE;
end
end
```

```
function r_interp = makeInterpolants(T,r_all,n)

r_interp = cell(n,1);

for i = 1:n
    ri = r_all(:,3*i-2:3*i);

    Fx = griddedInterpolant(T,ri(:,1),'spline');
    Fy = griddedInterpolant(T,ri(:,2),'spline');
    Fz = griddedInterpolant(T,ri(:,3),'spline');

    r_interp{i} = @(t) [Fx(t); Fy(t); Fz(t)];
end
end
```

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